Roll No. | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (2017 & Onward) (Sem.–3)

PHARMACEUTICAL ORGANIC CHEMISTRY-II

Subject Code : BP-301T

M.Code : 75105

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.**

SECTION-A

1. Answer briefly :

a) Define aromaticity.

b) Give the structure and use of BHC.

c) Why phenols are acidic in nature?

d) Give the basicity order of aniline, p-nitroaniline and m-nitroaniline.

e) Write any two reactions of cyclobutane.

f) What is Gatterman reaction?

g) Define ester value.

h) How many monosubstituted derivatives are possible for anthracene?

i) What is Birch reduction reaction?

j) "*Give the acidity order of benzoic acid, acetic acid and formic acid*". Justify.

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SECTION-B

2. Elaborate on various electrophilic substitution reactions of benzene by giving the detailed mechanism of any one. Discuss in detail the effect of substituents on the reactivity of these reactions.

3. Comment upon "*the stability of cycloalkanes with the help of various theories*".

4. Comment upon "*Haworth synthesis of naphthalene*". Mention various types of chemical reactions of naphthalene giving example of each category.

SECTION-C

5. Comment upon "*the synthetic evidences utilized for the derivation of structure of benzene"*.

6. Justify "*Chlorobenzene is less reactive but ortho and para directing towards electrohillic substitution reactions*".

7. Enumerate various qualitative tests carried out for the detection of phenols.

8. Discuss various reactions of aryl diazonium salts which involve retention of nitrogen.

9. What is saponification value? Give its significance.

10. Write a short note on the structure and medicinal uses of triphenylmethane.

11. Write a brief note on the Bayer's strain theory and its limitations.

12. Comment upon "*different reactions of fatty acids*".

13. Give the structure and uses of resorcinol and chloramine.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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